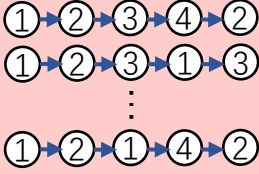


a

Random walk: $l = 5$ 

$$e_v = (f_v(\tilde{p}_1), f_v(\tilde{p}_2), \dots, f_v(\tilde{p}_M))$$



$$S^{(AP)} = \mathcal{M}^{(k)} \odot (\tilde{E}\tilde{E}^\top)$$

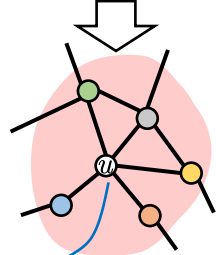
$$s.t. \text{dist}_A(u, v) \leq k$$

**Anonymous-Path-Driven
Structural Prior**

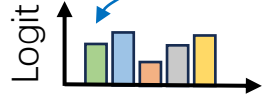


b

$$T = \text{softmax}(\text{SGC}(A, X; \Omega))$$



Node Class
● ● ● ● ●



$$S^{(CA)} = \mathcal{M}^{(k)} \odot (\tilde{T}\tilde{T}^\top)$$

$$s.t. \text{dist}_A(u, v) \leq k$$

**Class-Affinity-Driven
Semantic Prior**



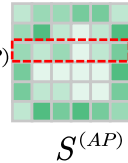
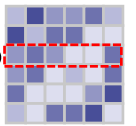
c

$$\omega_i^{(AP)} = \mathbf{q}^\top \tanh(W_{AP} S_i^{(AP)})$$

$$\omega_i^{(CA)} = \mathbf{q}^\top \tanh(W_{CA} S_i^{(CA)}) \quad S_i = \alpha_i^{(AP)} S_i^{(AP)} + \alpha_i^{(CA)} S_i^{(CA)}$$

$$\alpha_i^{(AP)}, \alpha_i^{(CA)} = \text{softmax}([\omega_i^{(AP)}, \omega_i^{(CA)}])$$

Pareto-Optimal Integration of $S^{(AP)}$ and $S^{(CA)}$


 $S^{(AP)}$

 $S^{(CA)}$


d

Optimization Objective:

$$\mathcal{L}_{\text{seesaw}} = L_{\text{GNN}_S} + L_{\text{GNN}_A}$$

Incremental Trick: $S^* \leftarrow S_{\text{new}}$

Masking Trick: $S_{\text{masked}} = S \odot \mathbb{I}_{\geq \tau}$

Graph Learning on the Balanced Graph Topology